Exercise 10 - Polar plots and the Nyquist condition

1 Polar plot - continued

If you remember from last week's exercise, in the polar plot we represented the frequency response $G(j\omega)$ of a system by plotting it in the complex plane as a function of ω.
There are no special rules to draw it, but the same principles as for the Bode plot apply: we start from $\omega = 0$ and increase it to $\omega \to \infty$.
Furthermore, as we saw, it is convenient to sktech a Bode plot first and then use it to draw the polar plot.
In addition, we can say that the two things that really matter in the polar plot are:

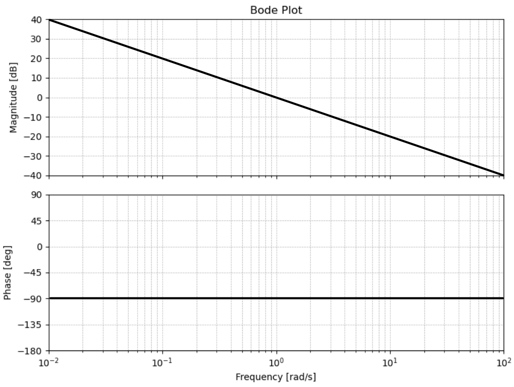
- Where the plot intersects the units circle (i.e. $|G(j\omega)| = 1$)
- Where the plot crosses the negative real axis (i.e. $\angle G(j\omega) = -180°$)

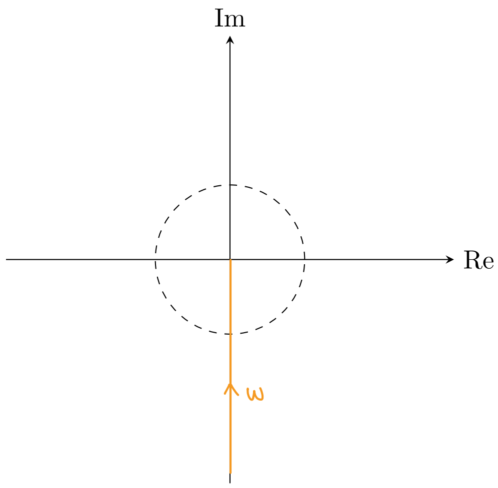
This because these two points are the ones that determine the stability of the closed-loop system, but we will see this in more detail in the following weeks.

Let's watch again some examples to refresh our memory about how to draw polar plots.

Example 1

Consider the following transfer function $G(s) = \frac{1}{s}$ with the following Bode plot:



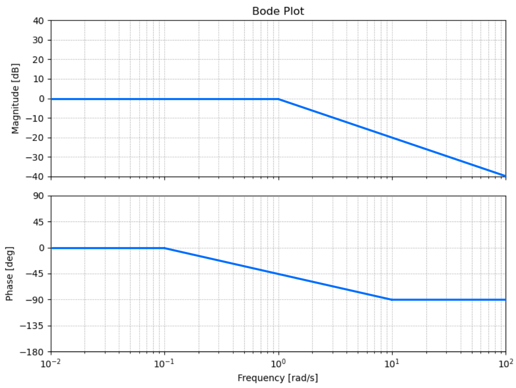


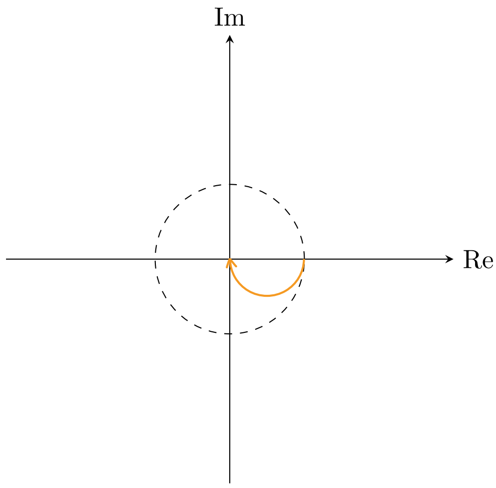
Example 2

Now if we add a pole at $s = -1$ we have

$$G(s) = \frac{1}{(s+1)}$$

with the following Bode plot:



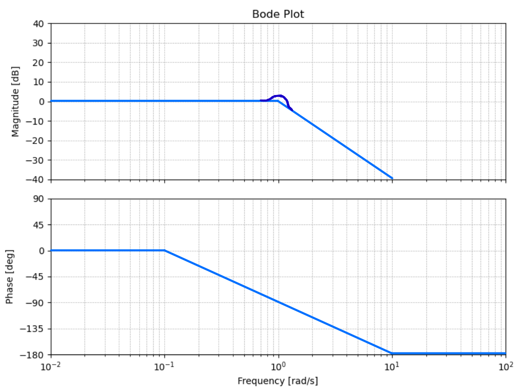


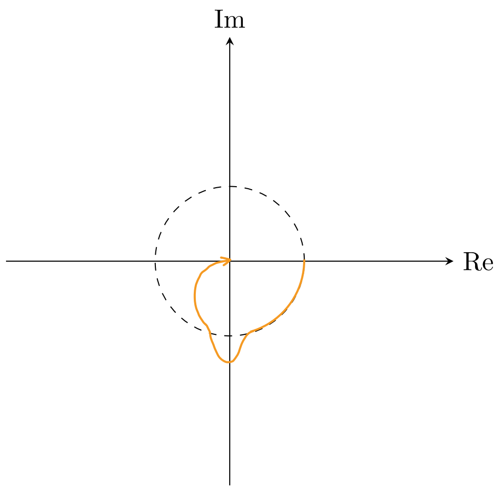
Example 3

For a more complex transfer function

$$G(s) = \frac{1}{s^2 + 2s + 1}$$

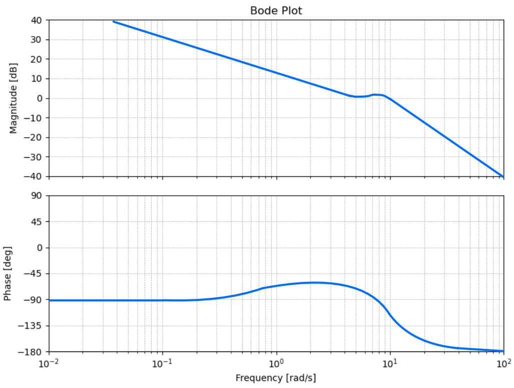
we get:

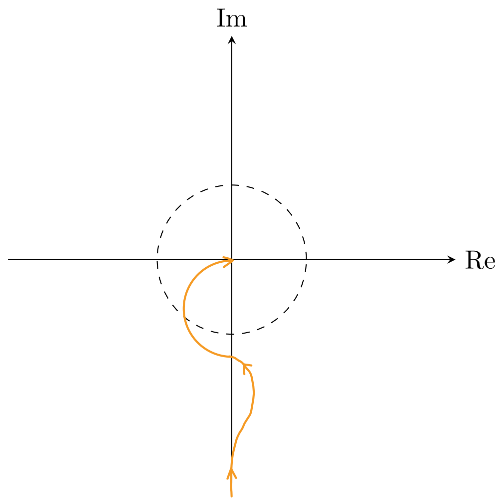




Example 4

Finally for the following Bode plot we have:





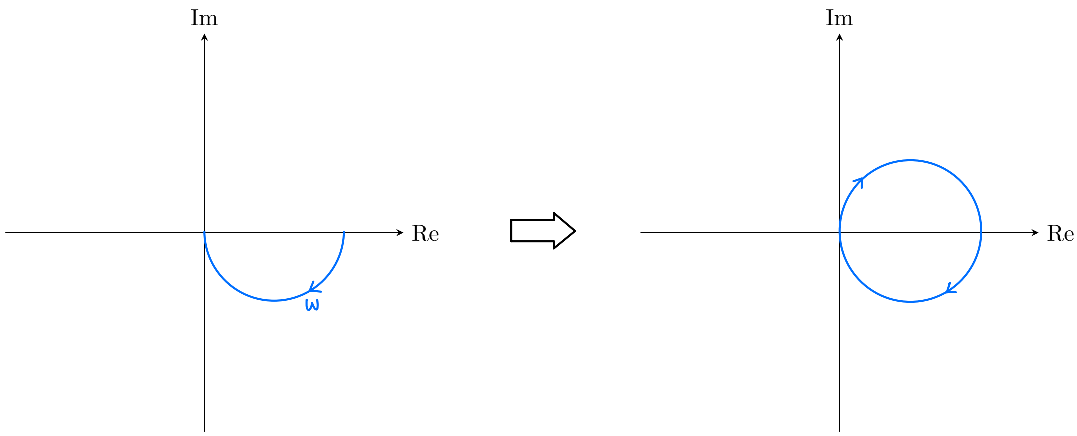
This are things that we already saw last week and that you should also have seen in the lecture.
The important concepts to remember from here are:

- Check where the plot starts (i.e. $\omega \to 0$)
- Check where the plot ends (i.e. $\omega \to \infty$)
- Check for key frequencies where the slope changes (i.e. poles and zeros)

The same applies for the Nyquist plot that we will see in the next section.

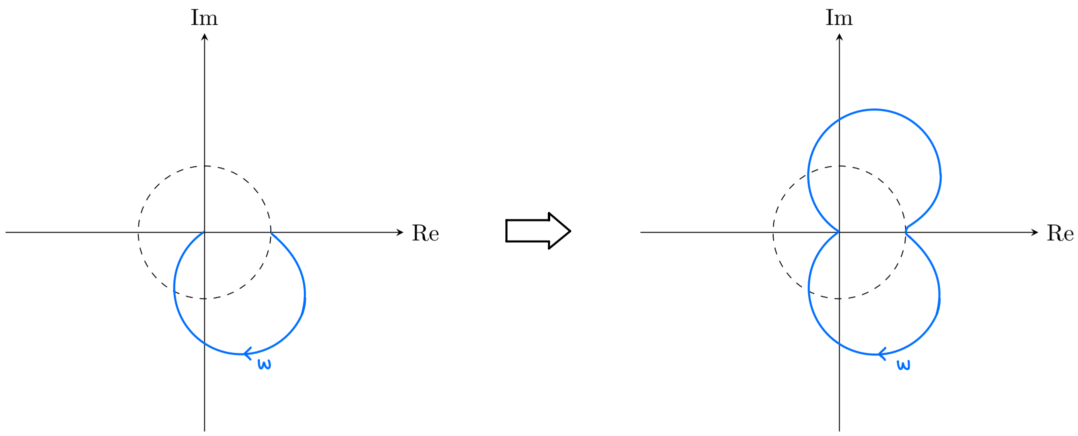
2 Nyquist plot

The Nyquist plot is very similar to the polar plot, in fact to draw it we simply take the polar plot and mirror it along the real axis.

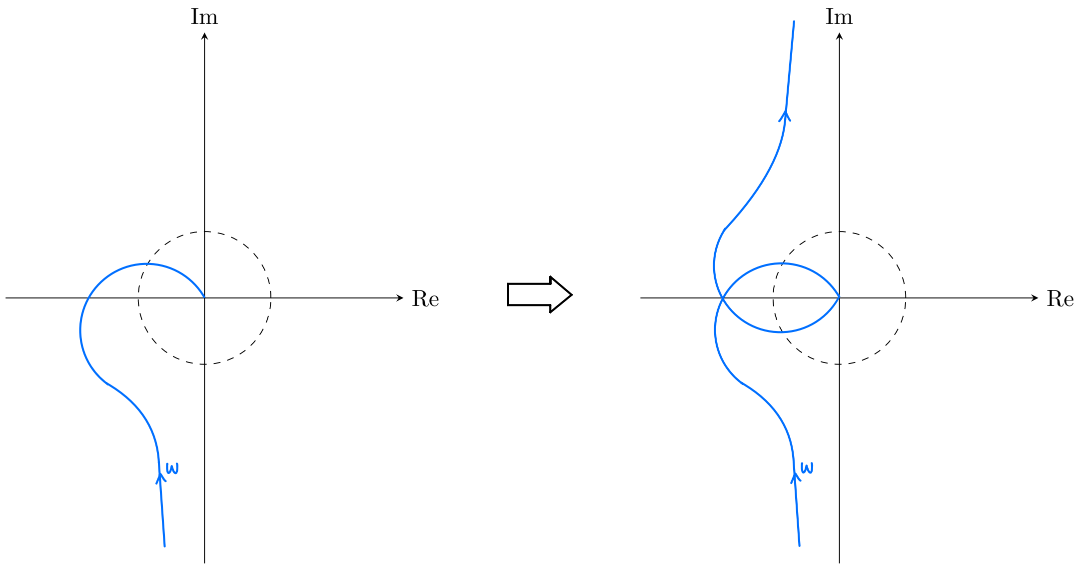


We will understand the reason for this in a bit, but first let's look at some more examples of Nyquist plots.

Example 1



Example 2



Thanks to the Nyquist plot we will be able to determine the closed-loop behavior of a system using only its open-loop description, i.e. to determine the stability and robustness of a closed-loop system.
This is done using the Nyquist stability criterion.

However, before seeing this in detail, let's take a step back and see how the Nyquist plot is derived. For that we need to take a quick look to complex analyisis and the so called principle of variation of the argument (we won't go in detail here and we won't prove anything, we will just look at the main idea).

2.1 Principle of variation of the argument

To gain some intuition about this principle, let's consider a transfer function

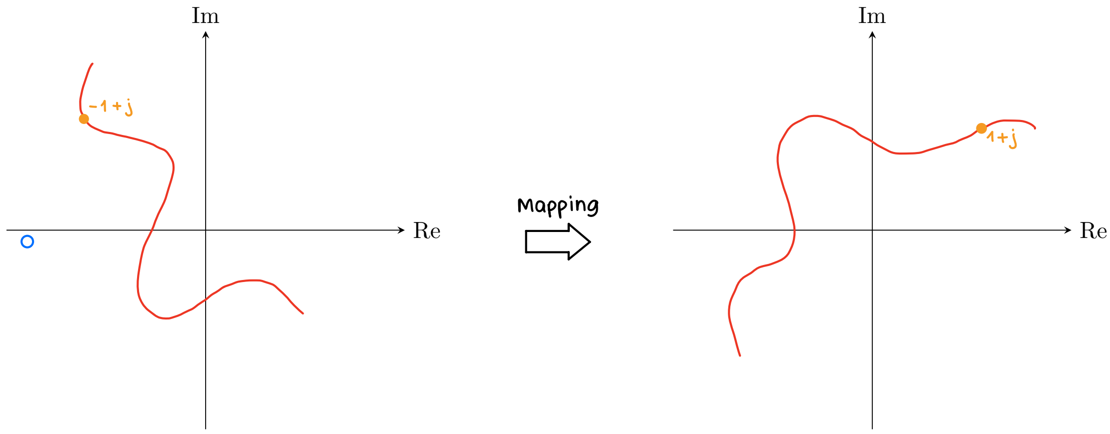
$$G(s) = s + 2$$

We want to see how the function $G(s)$ maps a certain contour in the complex plane, i.e. a closed curve.
As we already saw some weeks ago, the transfer function $G(s)$ can be depicted in the s-plane by plotting its poles (x) and zeros (o).

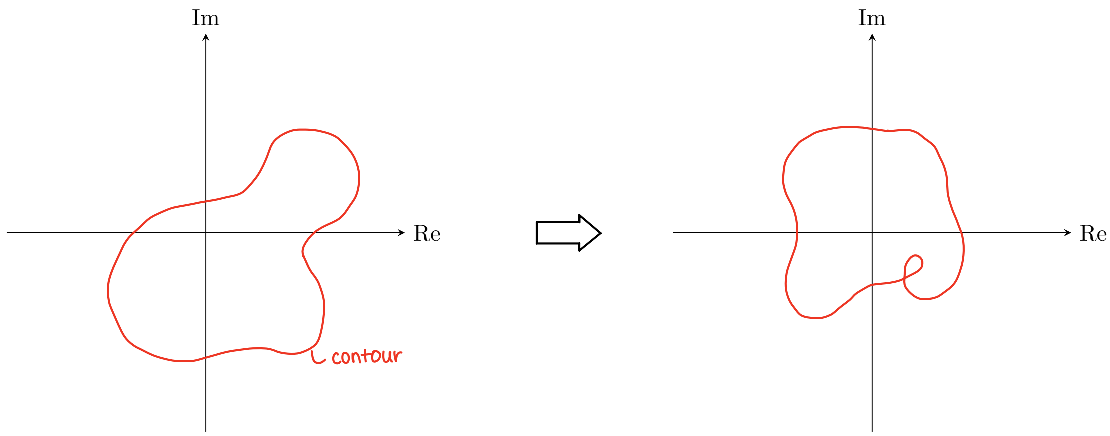
Let's now consider a random point $s_0 = -1 + j$ in the s-plane.
If we evaluate the function at that point we get:

$$G(s_0) = s_0 + 2 = 1 + j$$

This is another complex number.
We can now represent this new complex number in the complex plane, that we will call the $G(s)$-plane, since it contains the points obtained by evaluating $G(s)$ at different points in the s-plane.
We can extend this idea and map continuous lines in the s-plane to continuous lines in the $G(s)$-plane.

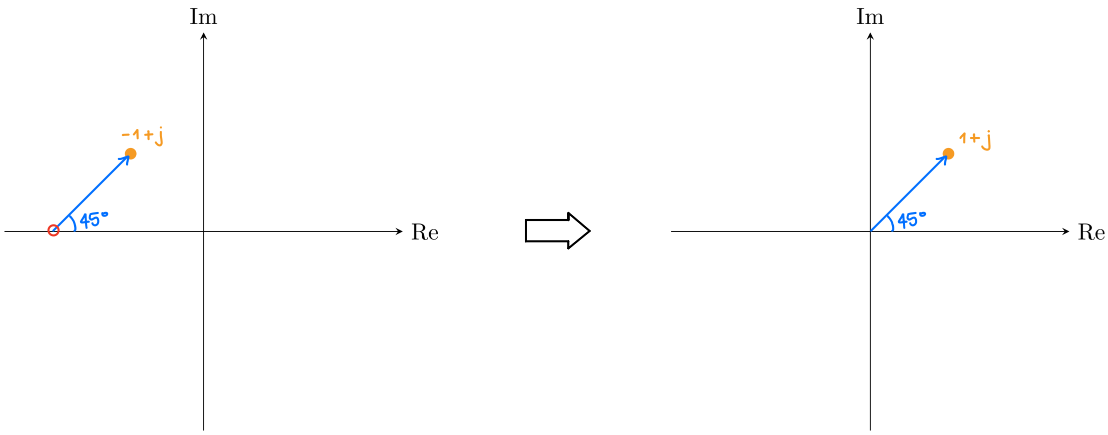


We are particarly interested in closed curves, i.e. curves that start and end at the same point. We will refer them as *contours*.



An important property is that a closed curve (contour) in the s-plane is mapped to a closed curve in the $G(s)$-plane, this is true for all contours in the s-plane.
We can also note that the closed curve in the $G(s)$-plane now also includes information about the transfer function $G(s)$ that we used to map the points.

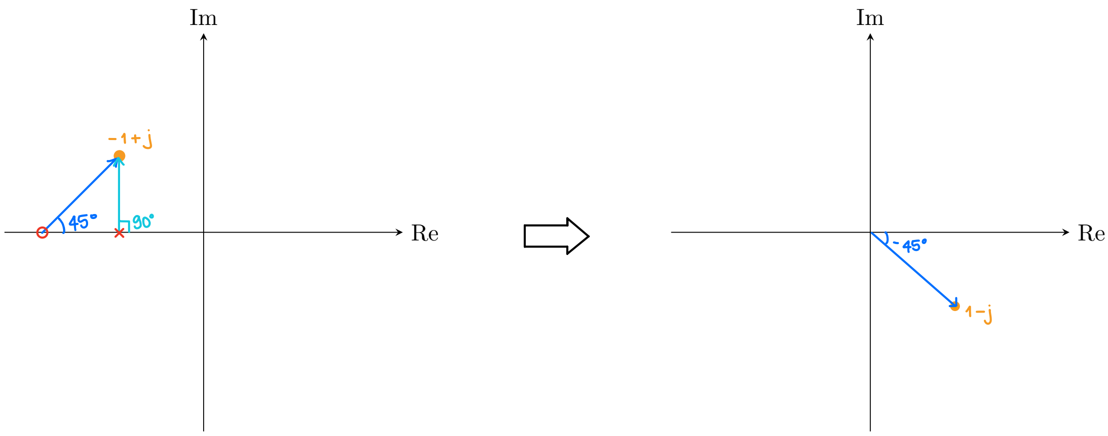
To see what information we can extract from this mapping, let's first consider the case of mapping one single point and then generalize it to contours.
Let's look at the example from before again.



Here we can see that the phase of the point in the $G(s)$-plane is the same as the phase from the zero to the point in the s-plane.
Now if we consider another transfer function where we have also a pole, e.g.

$$G(s) = \frac{s+2}{s+1}$$

we get the following mapping:



This happens becuase for a transfer function $G(s) = \frac{s-z_1}{s-p_1}$ the phase at a certain point s_0 is given by:

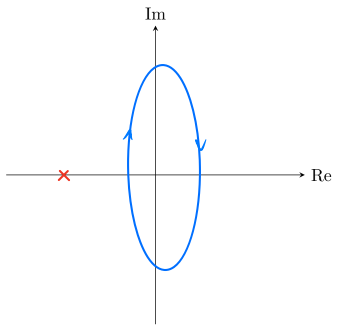
$$\angle G(s_0) = \angle(s_0 - z_1) - \angle(s_0 - p_1)$$

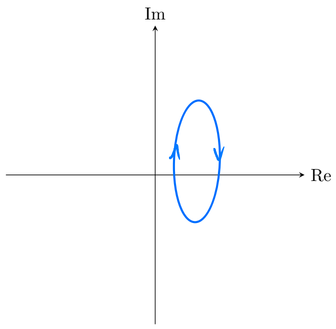
We can now generalize that the phase of a point mapped by some transfer function is given by the sum of the angles from the zeros to that point minus the sum of the angles from the poles to that point.

$$\angle G(s) = \sum_m \angle(s - z_i) - \sum_n \angle(s - p_n)$$

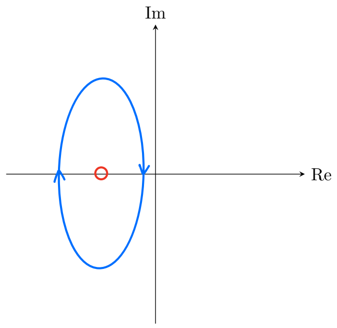
Now to see what happens when we map some closed contours, let's consider some examples.

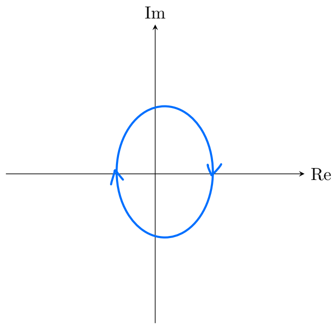
Example 1 - Contour not enclosing poles or zeros



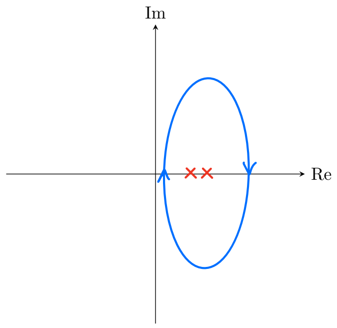


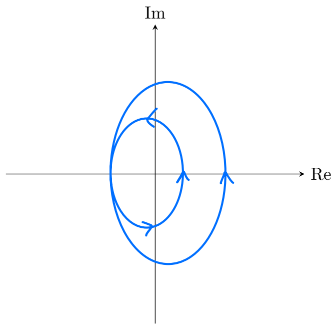
Example 2 - Contour enclosing a zero





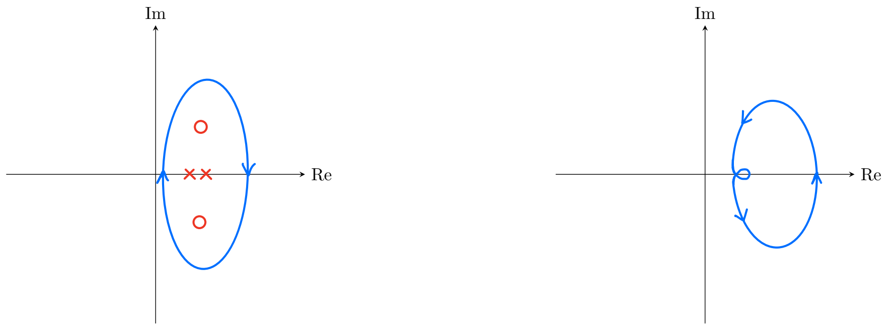
Example 3 - Contour enclosing two poles



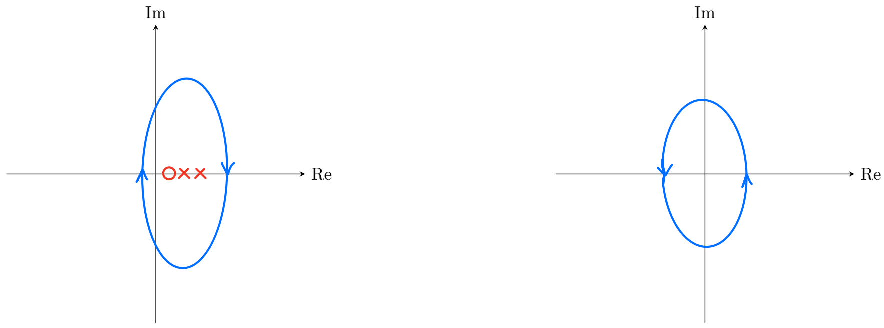


What we can see from these first examples is that when we enclose a zero or a pole in the contour in the s-plane, the mapped contour in the $G(s)$-plane also encircles the origin.
In particular, for each clockwise encirclement of a zero in the s-plane, we have a clockwise encirclement of the origin in the $G(s)$-plane.
Similarly, for each clockwise encirclement of a pole in the s-plane, we have a counter-clockwise encirclement of the origin in the $G(s)$-plane.
Note: You can think of it of a zero adding 360° and a pole subtracting 360°.

Example 4 - Contour enclosing two poles and two zeros



Example 5 - Contour enclosing two poles and one zero



Furthermore, from these last two examples we can see that if we have as many poles as zeros inside the contour in the s-plane, the mapped contour in the $G(s)$-plane does not encircle the origin.
On the other hand, if we have more poles than zeros inside the contour in the s-plane, the mapped contour encircles the origin in a counter-clockwise direction.

Theorem 6.1 (Variation of the Argument) *The number N of times that $G(s)$ encircles the origin of the complex plane as s moves along the contour Γ of a bounded simply-connected region D of the plane satisfies*

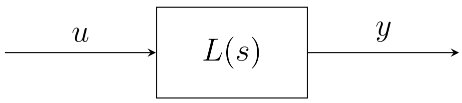
$$N = Z - P \tag{6.11}$$

where Z and P are the numbers of zeros and poles of $G(s)$ in D, respectively. Note that the encirclements are counted positive if in the same direction as s moves along Γ, and negative otherwise.

2.2 Nyquist stability criterion

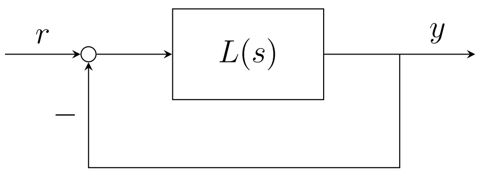
For now we have seen how with the principle of variation of the argument we can map contours in the s-plane to contours in the $G(s)$-plane and how the number of poles and zeros inside the contour in the s-plane affects the number of encirclements of the origin in the $G(s)$-plane.
But how is this relevant to us?
We can use this to determine the stability of a closed-loop system given its open-loop transfer function $G(s)H(s)$.

Recall that for an open-loop system



we can check its stability by looking at the poles of $L(s)$. If there are poles in the right half plane (RHP) the system is unstable, otherwise it is stable.

Now for a closed-loop system



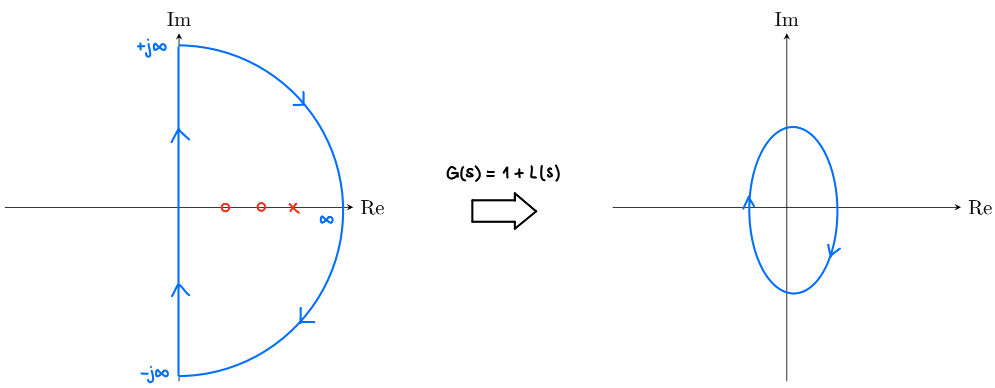
we have to look at the poles of the closed-loop transfer function

$$T(s) = \frac{L(s)}{1 + L(s)}$$

This means that we have to find all values of s such that

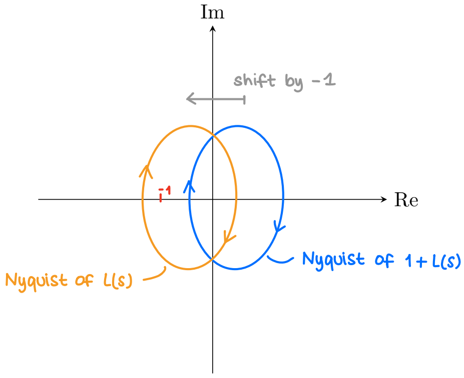
$$1 + L(s) = 0$$

This is equivalent to saying that to determine the stability of the closed-loop system we have to check if there are any zeros of $1 + L(s)$ in the RHP.
With this in mind we can now use the principle of variation of the argument introduced before and choose our D-contour, also called Nyquist contour, to encircle the entire RHP.
By doing this the mapping of this contour in the $G(s)$-plane will be equal to the Nyquist plot of $G(s)$.



Now, since we chose our contour to encircle the entire RHP, we can tell the relative difference of poles and zeros of $1 + L(s)$ in the RHP by looking at the number of encirclements around the origin the Nyquist plot of $1 + L(s)$.

Furthermore, we can note that the Nyquist plot of $1 + L(s)$ is simply the Nyquist plot of $L(s)$ shifted by 1 unit to the left.
This means that instead of looking at encirclements around the origin, we can equivalently look at encirclements around the point -1 in the Nyquist plot of $L(s)$.



This is good since we already know how to draw the Nyquist plot of $L(s)$.
To summarize, we can state that:

$$\#\text{Encirclements of } 0 \text{ using } 1 + L(s) =$$
$$= \#\text{Zeros of } 1 + L(s) \text{ in RHP} - \#\text{Poles of } 1 + L(s) \text{ in RHP} =$$
$$= \#\text{Encirclements of } (-1) \text{ using } L(s)$$

Now, let's take a look at the number of poles of $1 + L(s)$ in the RHP. How can we identify them?
Recall that the open-loop transfer function $L(s)$ is usually given as a ratio of polynomials

$$L(s) = \frac{N(s)}{D(s)}$$

The poles of $L(s)$ are given by the roots of the denominator $D(s)$.
Now, the poles of $1 + L(s)$ are the same as the poles of $L(s)$ since

$$1 + L(s) = 1 + \frac{N(s)}{D(s)} = \frac{D(s) + N(s)}{D(s)}$$

Thus, the number of poles of $1 + L(s)$ in the RHP is equal to the number of poles of $L(s)$ in the RHP, that are the unstable open-loop poles!

Now we can rewrite the previous statement as:

$$\#\text{Encirclements of } (-1) \text{ using } L(s) =$$
$$= \#\text{Zeros of } 1 + L(s) \text{ in RHP} - \#\text{Poles of } L(s) \text{in RHP}$$

Finally, we can rearrange this equation to get the Nyquist stability criterion:

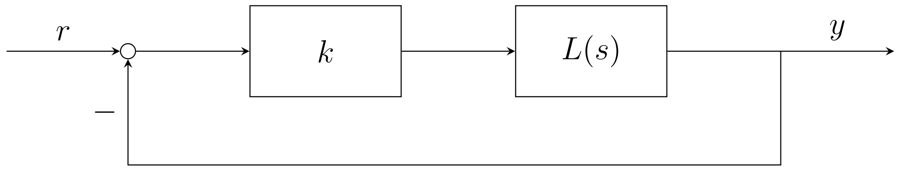
$$\boxed{Z = N + P}$$

where

- $Z \implies$ # zeros of $1 + L(s)$ in the RHP = # unstable closed-loop poles
- $N \implies$ # clockwise encirclements of the point -1 in the Nyquist plot of $L(s)$
- $P \implies$ # poles of $L(s)$ in the RHP = # unstable open-loop poles

This criterion allows us to determine the stability of the closed-loop system by simply looking at the Nyquist plot of the open-loop TF $L(s)$ and its unstable poles.

Note for proportional controller k:

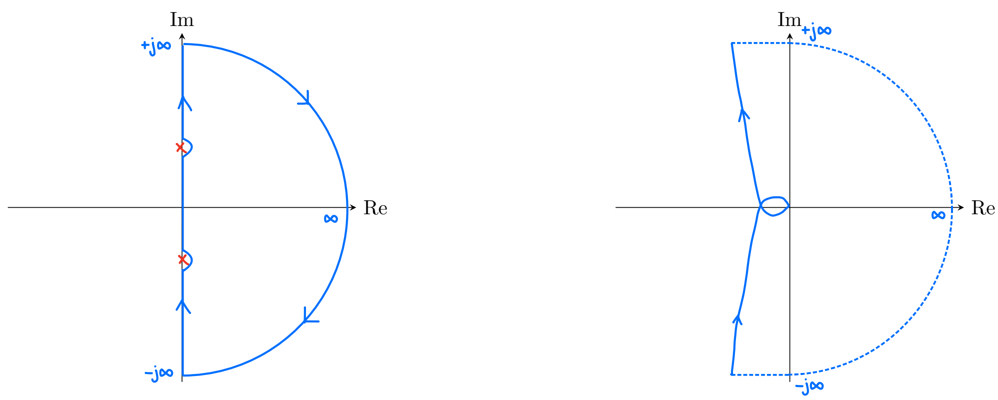


If we include a proportional feedback gain k in the system, our TF becomes

$$T(s) = \frac{kL(s)}{1 + kL(s)} = \frac{L(s)}{\frac{1}{k} + L(s)}$$

This means that to determine the stability of the closed-loop system we have to check the number of encirclements of the point $-\frac{1}{k}$ in the Nyquist plot of $L(s)$.
This is important because by changing the gain k we change the point we have to check for encirclements.
Thus, by increasing or decreasing k we can make the closed-loop system stable or unstable, by "changing" the number of unstable closed-loop poles.

Special case - poles on the imaginary axis:
If there are poles or zeros of $1 + L(s)$ on the imaginary axis, we cannot directly apply the Nyquist stability criterion.



In this case we have to slightly modify the Nyquist contour to avoid these poles. This is usually done by adding small semicircles around these points that go into the RHP. If you move around these poles CCW, you have to close the Nyquist-plot CW at infinity.

Exam Problem - HS22

Problem: Consider the feedback system depicted in Figure 16 below.

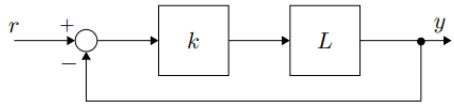


Figure 16: Standard feedback architecture. LTI System L and proportional gain k.

Let the transfer function of the linear time-invariant system L be given by

$$L(s) = 10 \cdot \frac{(s-8)(s-5)}{(s+6)(s+1)(s+10)}$$

The Nyquist plot of $L(s)$ is shown in Figure 17.

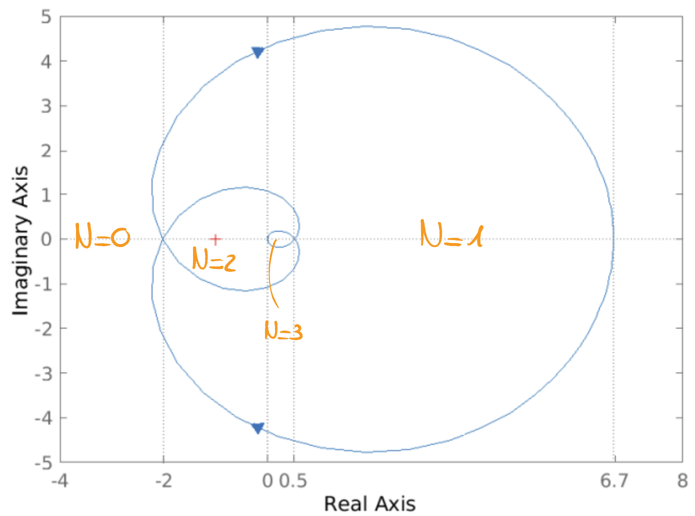


Figure 17: Nyquist plot of $L(s)$.

We now assess the number of closed-loop unstable poles for different values of k.

Q31 **(0.5 Points)** For $-2 < \frac{-1}{k} < 0$ the number of closed-loop unstable poles Z is

$Z = N + \cancel{P}_{0} = 2$

Q32 **(0.5 Points)** For $0 < \frac{-1}{k} < 0.5$ the number of closed-loop unstable poles Z is

$Z = N + \cancel{P}_{0} = 3$

Q33 **(0.5 Points)** For $0.5 < \frac{-1}{k} < 6.7$ the number of closed-loop unstable poles Z is

$Z = N + \cancel{P}_{0} = 1$

We know that the Nyquist stability criterion says that

$\Rightarrow Z = N + P$

where Z is the number of closed-loop unstable poles

N is the number of CW encirclements of $-\frac{1}{k}$

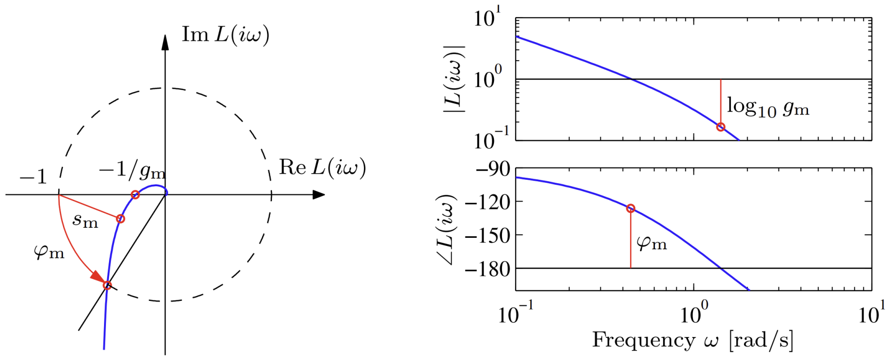
P is the number of open-loop unstable poles

$\Rightarrow$ From the TF $L(s)$ we see that it doesn't have unstable poles $\Rightarrow P = 0$

$\Rightarrow Z = N$

2.3 Stability Margins

The Nyquist plot can also be used to determine how far away a system is from instability.
If we assume our open-loop to be stable, i.e. $P = 0$, for our closed loop to be stable we need to have $N = 0$ and thus $Z = 0$.
This means that the Nyquist plot of $L(s)$ must not encircle the point -1.
We can now define two stability margins:



- **Gain margin g_m:** The point at 180°. Tells us how much we can increase the gain of $L(s)$ before the closed-loop system becomes unstable.
  This is given by the distance between the Nyquist plot and the point -1 along the negative real axis.

- **Phase margin φ_m:** The point at magnitude 1. It tells us how much we can change the phase until the closed-loop system becomes unstable.
  This is given by the angle between the negative real axis and the line connecting the origin to the point where the Nyquist plot intersects the unit circle.

Note: This margins are easier to see in the Bode plot as you can see from the figure above.
Remember that this special condition is only valid if the open-loop system is stable! Otherwise, we have to use the full Nyquist stability criterion or the Root Locus method.

Exam Problem - FS24

Problem: Let L be a minimum-phase and stable linear time-invariant system. The Bode plot for L is given in Figure 9. The open-loop system L is shown in Figure 10a, and the closed-loop system comprising L as shown in Figure 10b is referred to as T.

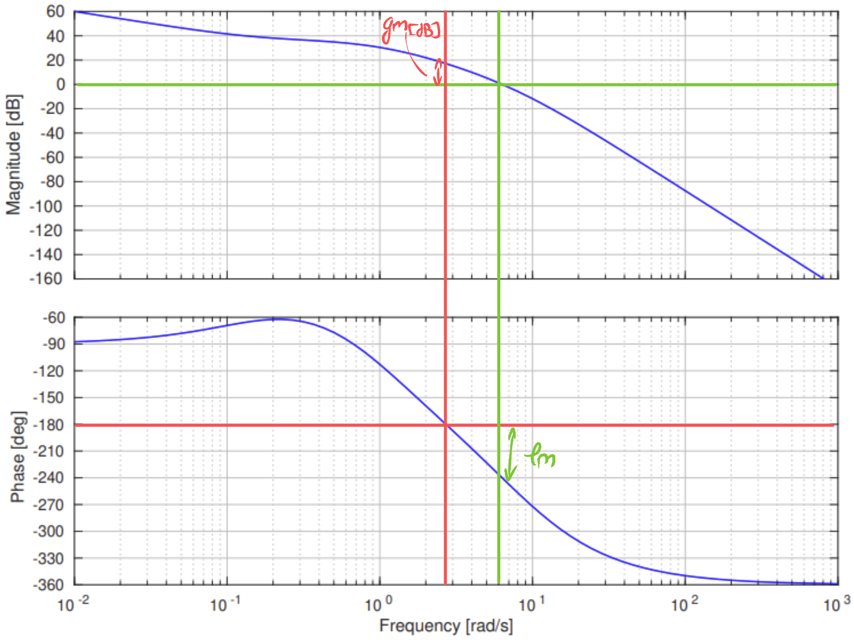
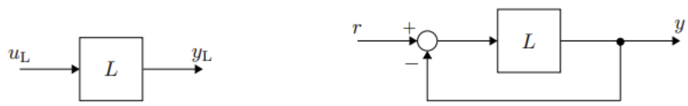


Figure 9: Bode plot of L.



(a) Open-loop system L. (b) Closed-loop system T.

Figure 10: Open-loop system L and closed-loop system T.

Q26 **(0.5 Points)** Mark the correct answer.
The phase margin PM for closed-loop stability is best described by,

- A $PM \approx 58°$
- B $PM \approx -61°$
- C $PM \approx 34°$
- D $PM \approx -26°$

Q27 **(0.5 Points)** Mark the correct answer.
The gain margin GM for closed-loop stability is best described by,

- A $GM \approx 18$ dB
- B $GM \approx 5$ dB
- C $GM \approx 90$ dB
- D $GM \approx -17$ dB

- As you can see from the picture above we can identify the PM and the GM by drawing an horizontal line at −180° and another one at 0dB.
- Then we draw a vertical line that that passes through the point where the phase/gain plot intersects the horizontal line ⇒ the new distance is therefore the gain and the phase margin respectively
- PM is defined positive going up
  ⇒ $PM = -241° + 180° = -61°$
- GM is defined positive going down
  ⇒ $GM = 0dB - 17dB = -17dB$